

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Crankshaft Seal Replacer Kit, Part Number 5394206, or equivalent
- Lubriplate™ 105, Part Number 3163086, or equivalent

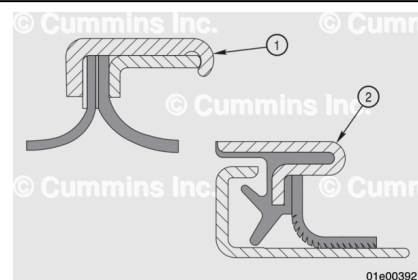
Additional Service Items

- Magnet
- Crocus cloth
- Clean, lint-free cloth

General Information

There are two types of seals used on this engine: a wet oil seal (1) and a dry oil seal (2). The wet oil seal has a wear sleeve. The dry oil seal does **not** have a wear sleeve.

The wet oil seal (1) is used in engines with wet type flywheel housings. The dry oil seal (2) is used in engines with dry type flywheel housings.



Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

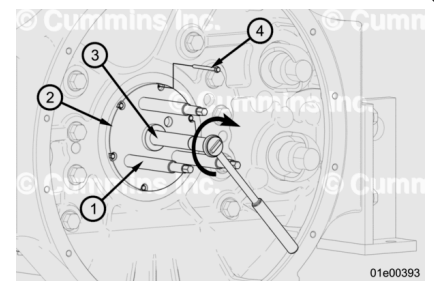
This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Disconnect the batteries. See equipment manufacturer service information.
- Remove the transmission, clutch, and all related components. See equipment manufacturer service information
- Remove the engine speed sensor. See equipment manufacturer service information
- Remove the flywheel. Refer to Procedure 016-005 in Section 16.
(/qs3/pubsys2/xml/en/procedures/89/89-016-005.html)

Remove

⚠ CAUTION ⚠

Do not use an impact wrench or air tools. Doing so can damage the tool.



Remove the rear crankshaft oil seal. Use Crankshaft Seal Replacer Kit, Part Number 5394206, or equivalent.

Apply Lubriplate™ 105, Part Number 3163086, or equivalent, onto the remover screw threads and tip (3).

Install the three replacer screws (1) into the crankshaft.

Install the remover screw (3) into the replacer plate (2).

Slide the replacer plate assembly (2) into the nose of the crankshaft.

Drill six self-tapping metal screws (4) into the rear oil seal. Make sure the heads of the screws are flush on the face of the tool. Six screws **must** be used in this process to make sure the seal will come out evenly.

Use a wrench to rotate the remover screw (3) **clockwise** against the crankshaft rear boss slowly, which slides the replacer plate (2) along with the seals outward.

Once the rear oil seal and dust seal are removed from the crankshaft, remove and discard the self-tapping metal screws (4) to separate the rear oil seal from the crankshaft seal replacer assembly (2).

Remove the three replacer screws (1) from the crankshaft.

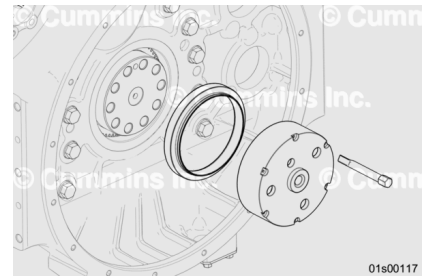
Discard the oil seal.

The metal wear sleeve will still be on the crankshaft if the engine uses the wet oil seal.

Use the wear sleeve expander and replacer plate to remove the wear sleeve from the crankshaft.

Rotate the wear expander in the replacer plate to remove the wear sleeve.

Discard the old sleeve.



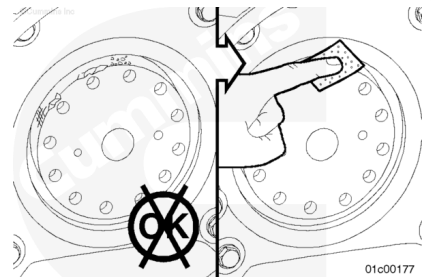
Clean and Inspect for Reuse

Use a magnet to remove any metal chips from the crankshaft and crankshaft seal bore.

Clean the crankshaft and seal mounting area.

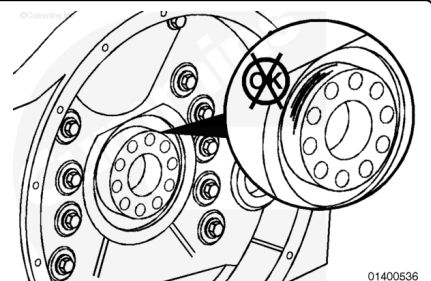
Use a crocus cloth to polish and remove deposits from the crankshaft. Visible scratches on the sealing surface can result in seal leakage.

Use a clean, lint-free cloth to clean the sealing surface of the crankshaft and remove all traces of oil film from the crankshaft.



Inspect the crankshaft end for nicks, burrs, and grooves.

If grooves on the crankshaft sealing surface exceed 10µm in depth, a wear sleeve **must** be used.



Install

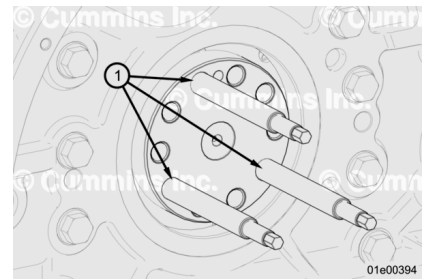
Wet

⚠ CAUTION ⚠

Do not lubricate. The seal, crankshaft, and flywheel housing must be clean and dry or a leak can result.

⚠ CAUTION ⚠

Do not remove the seal from the wear sleeve. If the seal is removed from the wear sleeve, it can be damaged during the installation process.

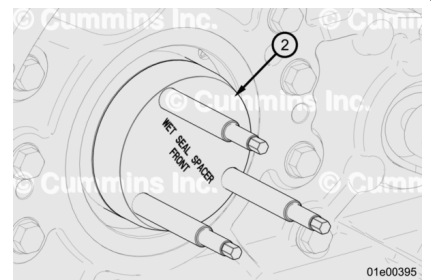


Use Crankshaft Seal Replacer Kit, Part Number 5394206, or equivalent to install the rear crankshaft oil seal.

Apply Lubriplate™ 105, Part Number 3163086, or equivalent, onto the installer screw threads and tip.

Install the three replacer screws (1) in the crankshaft.

Slide the crankshaft seal guide (2) until it touches the crankshaft face. Use the crankshaft seal guide that has the "Wet Seal Spacer Front" marking. The marking **must** face outward.



Note : The wet oil seal part number is printed on the front face of the seal. The part number **must** face away from the cylinder block once installed.

Note : The wear sleeve and seal **must** be installed simultaneously using a special tool. Attempting to install the assembly without the tool will result in the malfunction of the seal and an oil leak.

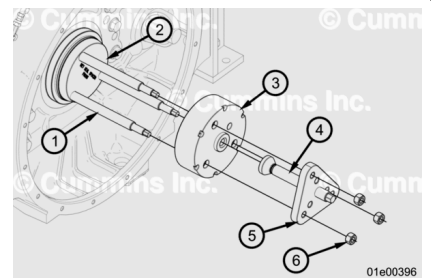
Face the oil seal part number outward and slide over the crankshaft seal guide nose (2). Make sure the seal is positioned squarely on the crankshaft seal guide.

The replacer plate (3) is **only** used for the installation of the rear wet seal. Do **not** install the replacer spacer (7) into the replacer plate (3).

Slide the replacer plate (3) up to the nose of the crankshaft seal guide (2).

Install the replacer plate (5) and installer screw onto the replacer screws (1).

Install the three regular hexagon jam nuts (6).



Move the crankshaft seal replacer (2) toward the rear seal by rotating the installer screw (4) **clockwise** until it is positioned against the oil seal.

Holding the crankshaft seal replacer (2), rotate the installer screw (3) **clockwise** until the crankshaft seal replacer is positioned against the crankshaft seal guide.

Remove the crankshaft seal replacer (2) and crankshaft seal guide (2) from the replacer screw assembly.

Dry

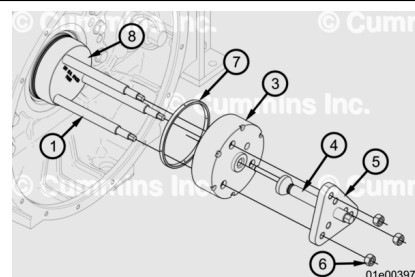
Note : The dry oil seal part number is printed on the front face of the seal. The part number must face away from the cylinder block once installed.

Slide the crankshaft seal guide (8) to touch the crankshaft face. Use the crankshaft seal guide (8) that has a "Dry Seal Spacer Front" marking. The marking should face the operator.

Face the dry oil seal part number outward and slide over the crankshaft seal guide (8) nose. Make sure the seal is positioned squarely on the crankshaft seal guide.

The replacer plate (3) and replacer spacer (7) assembly is used to install the rear dry seal. Install the replacer spacer (7) into the replacer plate (3) before sliding the replacer plate assembly.

Repeat the steps in the Install - Wet section.

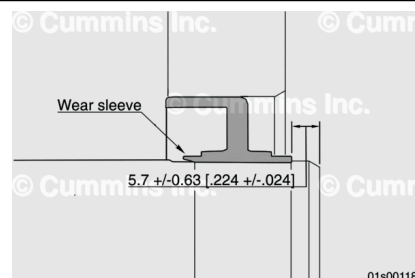


Measure

Wet

Measure the rear wet seal sleeve installation depth from the crankshaft mounting face.

Rear Wet Seal Sleeve Installation Depth from Crankshaft Mounting Face		
mm		in
5.07	MIN	0.2
6.33	MAX	0.249

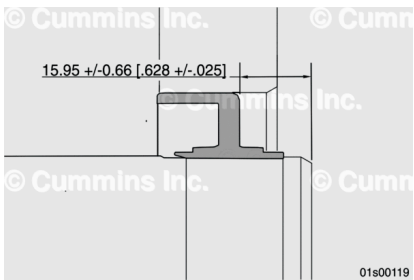


If the rear wet sleeve installation depth is **not** within specifications, the rear wet seal **must** be replaced.

Measure the rear wet seal installation depth from the crankshaft mounting face.

Rear Wet Seal Installation Depth from Crankshaft Mounting Face

mm		in
15.29	MIN	0.602
16.61	MAX	0.654



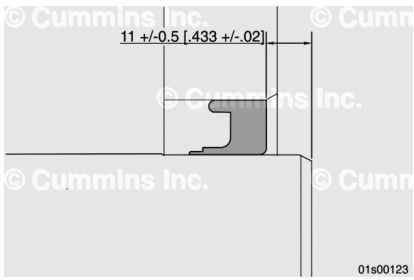
If the rear wet seal installation depth is **not** within specifications, the rear wet seal **must** be replaced.

Dry

Measure the rear dry seal installation depth from the crankshaft mounting face.

Rear Dry Seal Installation Depth from Crankshaft Mounting Face

mm		in
10.5	MIN	0.413
11.5	MAX	0.453



If the rear dry seal installation depth is **not** within specifications, the rear dry seal **must** be replaced.

Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Install the flywheel. Refer to Procedure 016-005 in Section 16. (/qs3/pubsys2/xml/en/procedures/89/89-016-005.html)
- Install the engine speed sensor. See equipment manufacturer service information.
- Install the transmission, clutch, and all related components. See equipment manufacturer service information.
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine. Check for leaks.